

Sequential Decision Analysis for Fleet Replacement in Electric Truck Adoption

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with supervision from

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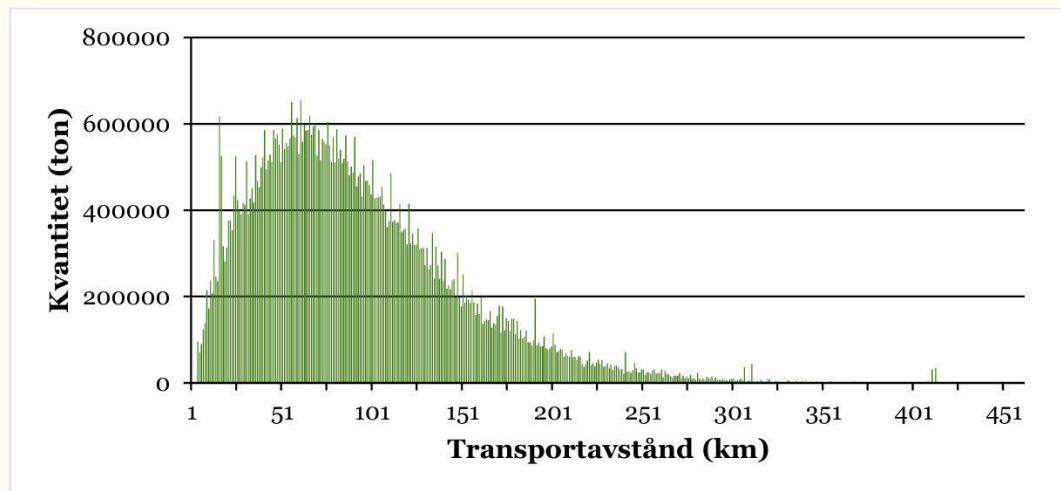
Agenda

- Swedish forestry transports
- Fleet replacement problem
- Simulation model walkthrough
- Results
- Conclusions & Future work

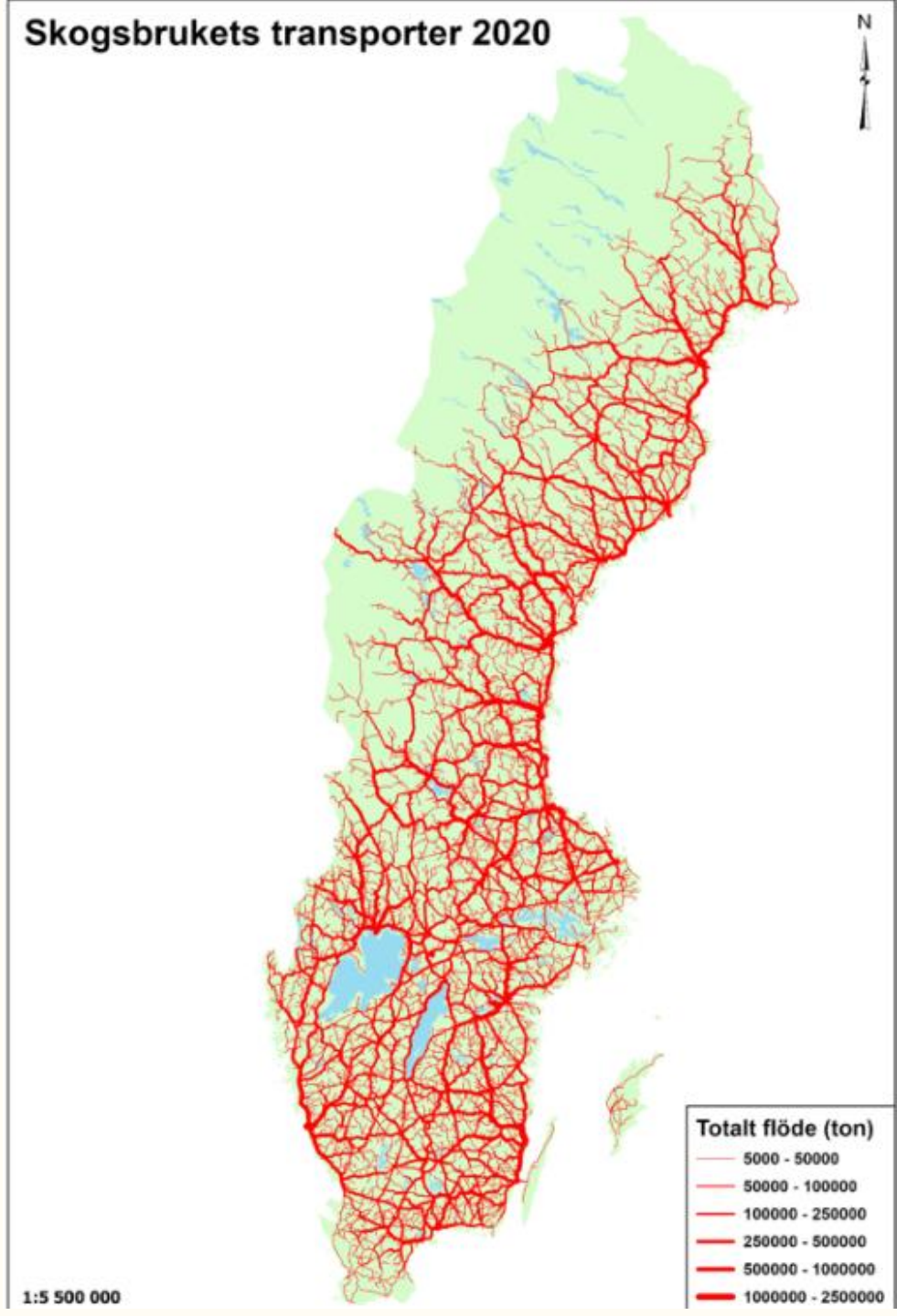


Forestry transports in Sweden

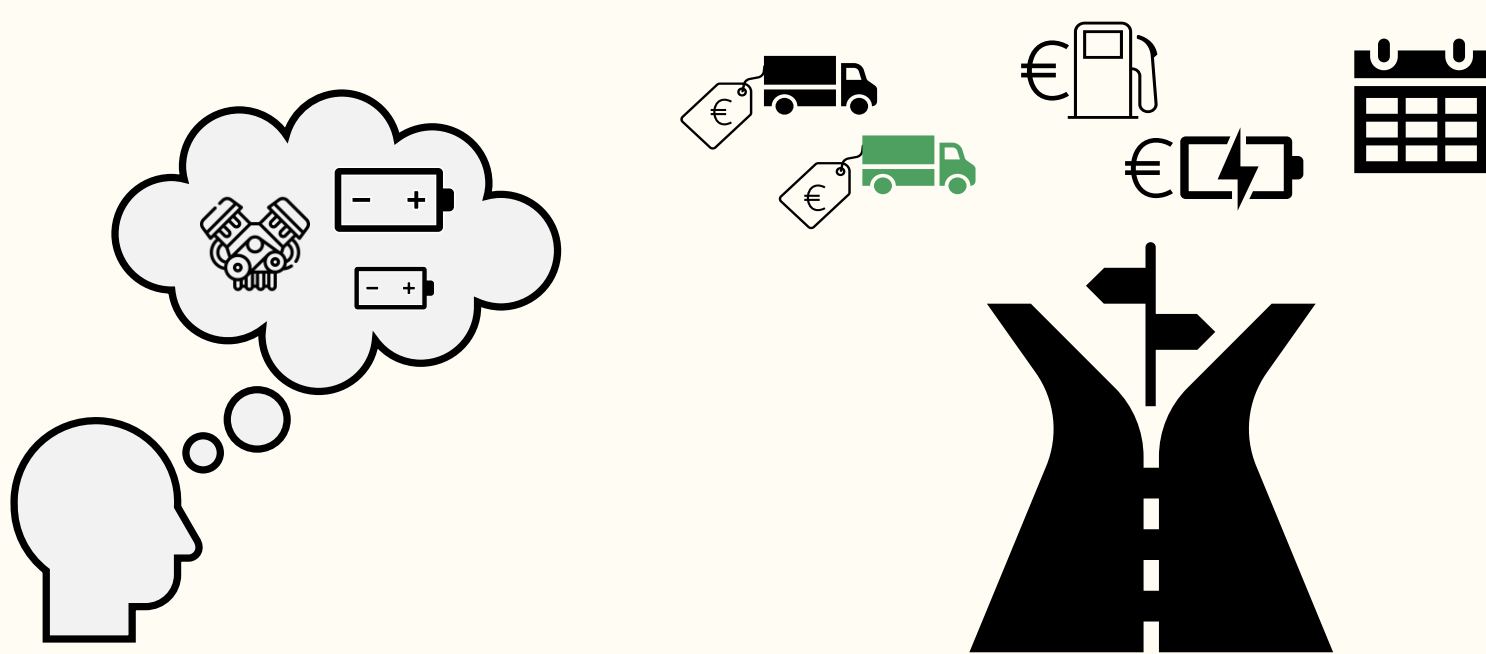
- Near 20% of Swedens heavy road transports
- 2 million transports each year
- From ~300 000 pickup locations in the forest
- To ~2000 receiving industries
- 64 - 74 tons
- Average transport distance: 90 km
- 50% of forestry sector's CO₂
- 30% of forestry sectors's costs
- Goal: Fossil-free by 2040



Skogsbrukets transporter 2020



Fleet owners' vehicle replacement decision



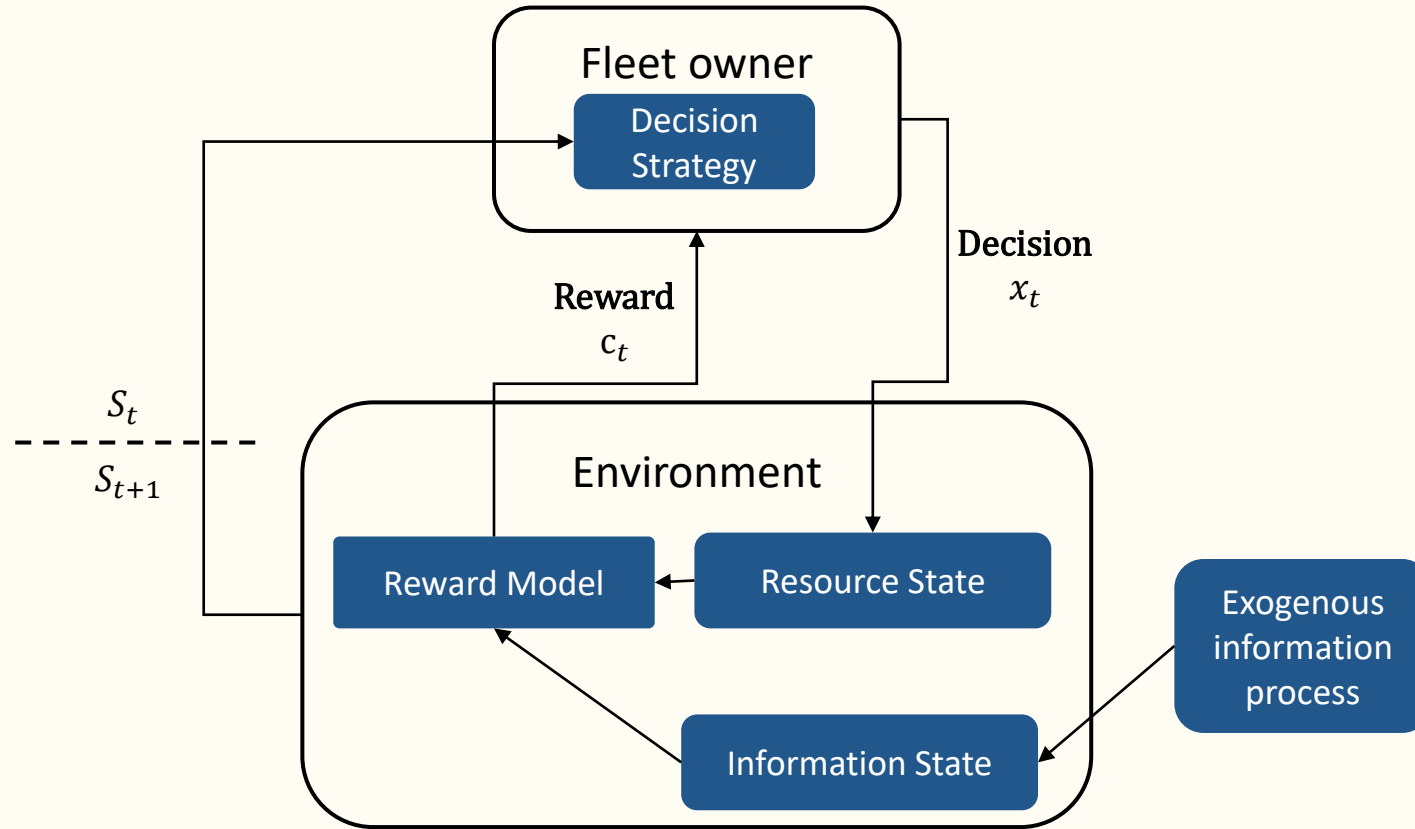
- Optimal timing of replacement
- Competing technologies
- Risks with investing early
 - Premature technology
 - High investment cost
- Risks with waiting
 - Loss of residual value
- Use case
 - Decision support for fleet owner
 - Analysis tool for policy makers

2026

2045

Fleet replacement

A sequential decision making process



$$\max_{\pi} \mathbb{E} \left\{ \sum_{t=0}^T \gamma^t C(S_t, X^{\pi}(S_t)) \mid S_0 \right\}$$

Resource state and decision

$$R_t = (\text{DT}, \text{BET}, \dots, \text{DT}, \text{BET})$$

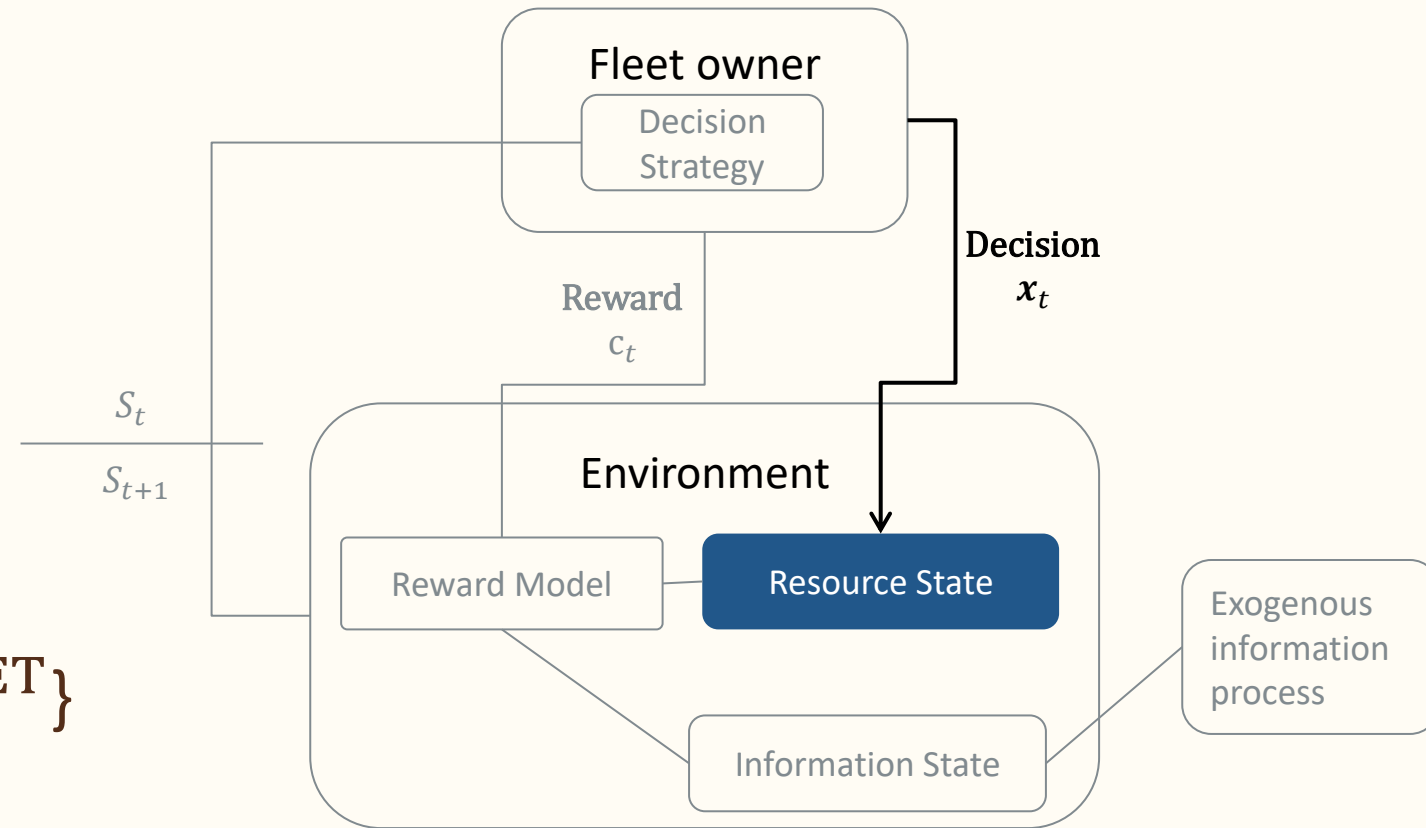





Type (DT, BET) age purchase price

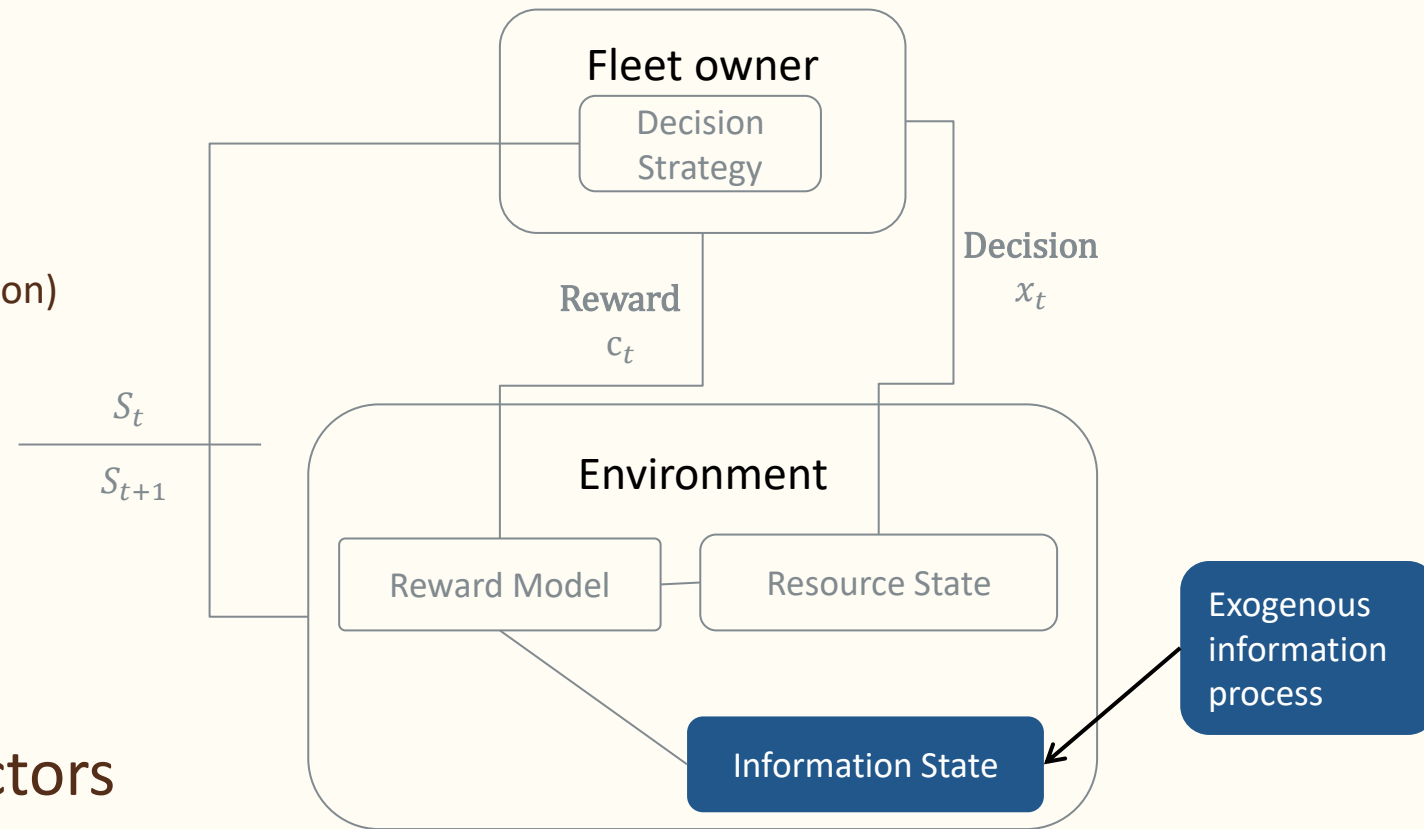
$$\mathbf{x}_t = (x_t^1, x_t^2, \dots, x_t^n)$$

$$x_t^i \in \{\text{keep}, \text{replace}^{\text{DT}}, \text{replace}^{\text{BET}}\}$$



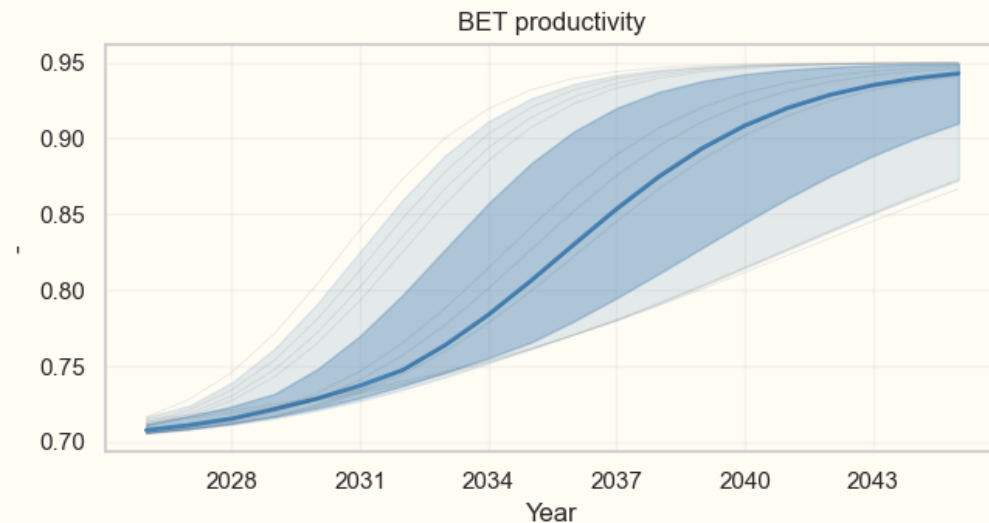
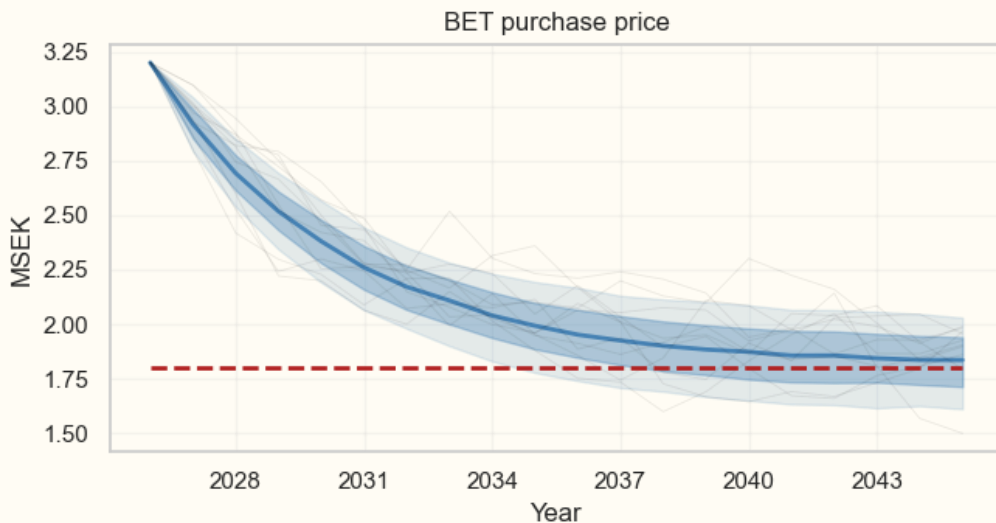
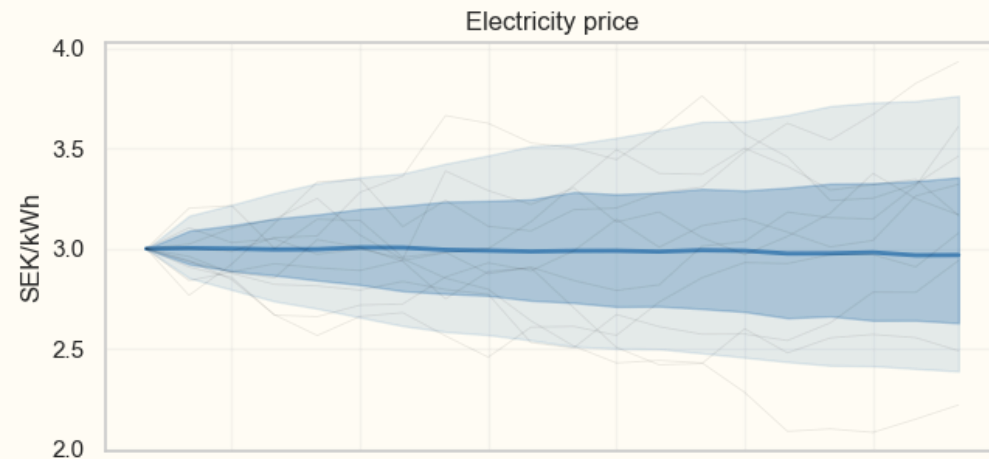
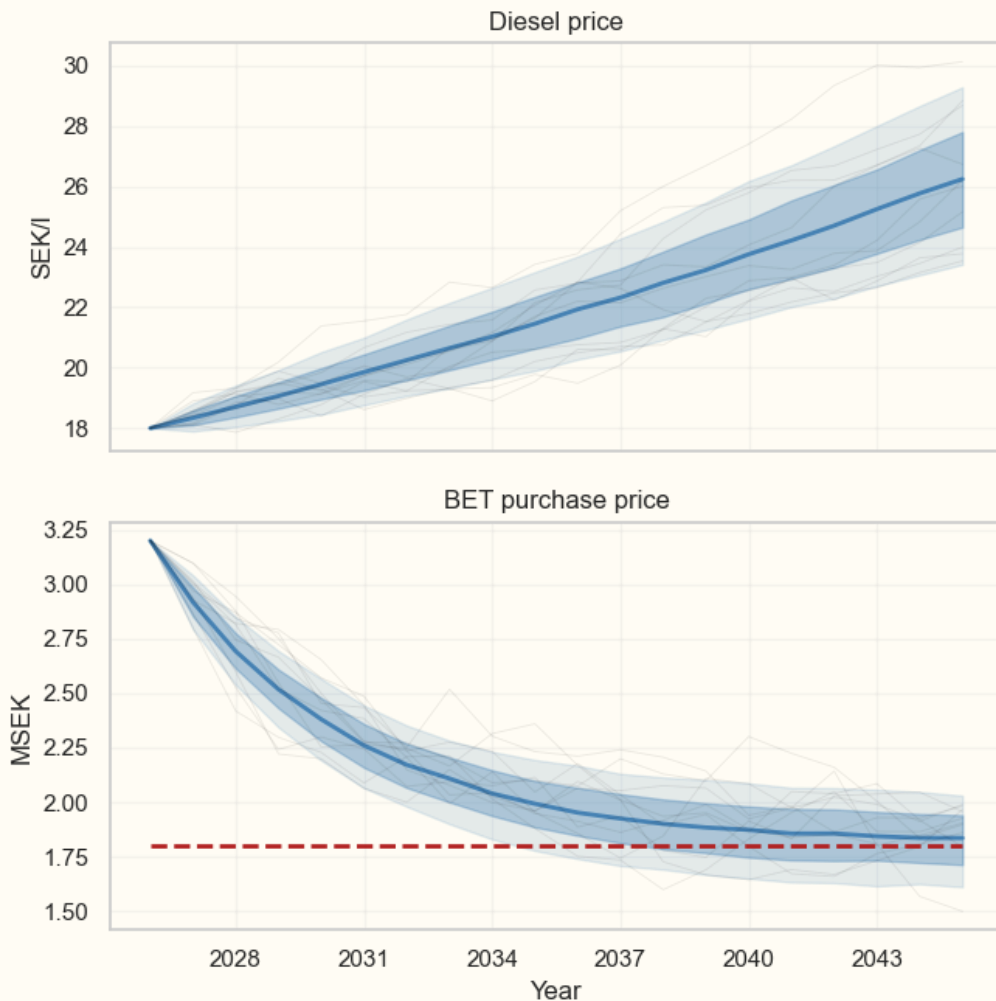
Exogenous information

- Energy price
 - Diesel - (Geometric Brownian Motion)
 - Electricity - (Geometric Brownian Motion)
- Purchase price
 - DT - (Constant)
 - BET - (Ornstein-Uhlenbeck)
- BET Productivity - (S-curve)
 - Vehicle and infrastructure factors



Uncertainty modelling

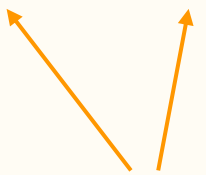
10-90 percentile
 25-75 percentile
 Median
 Sample pathways
 DT Price



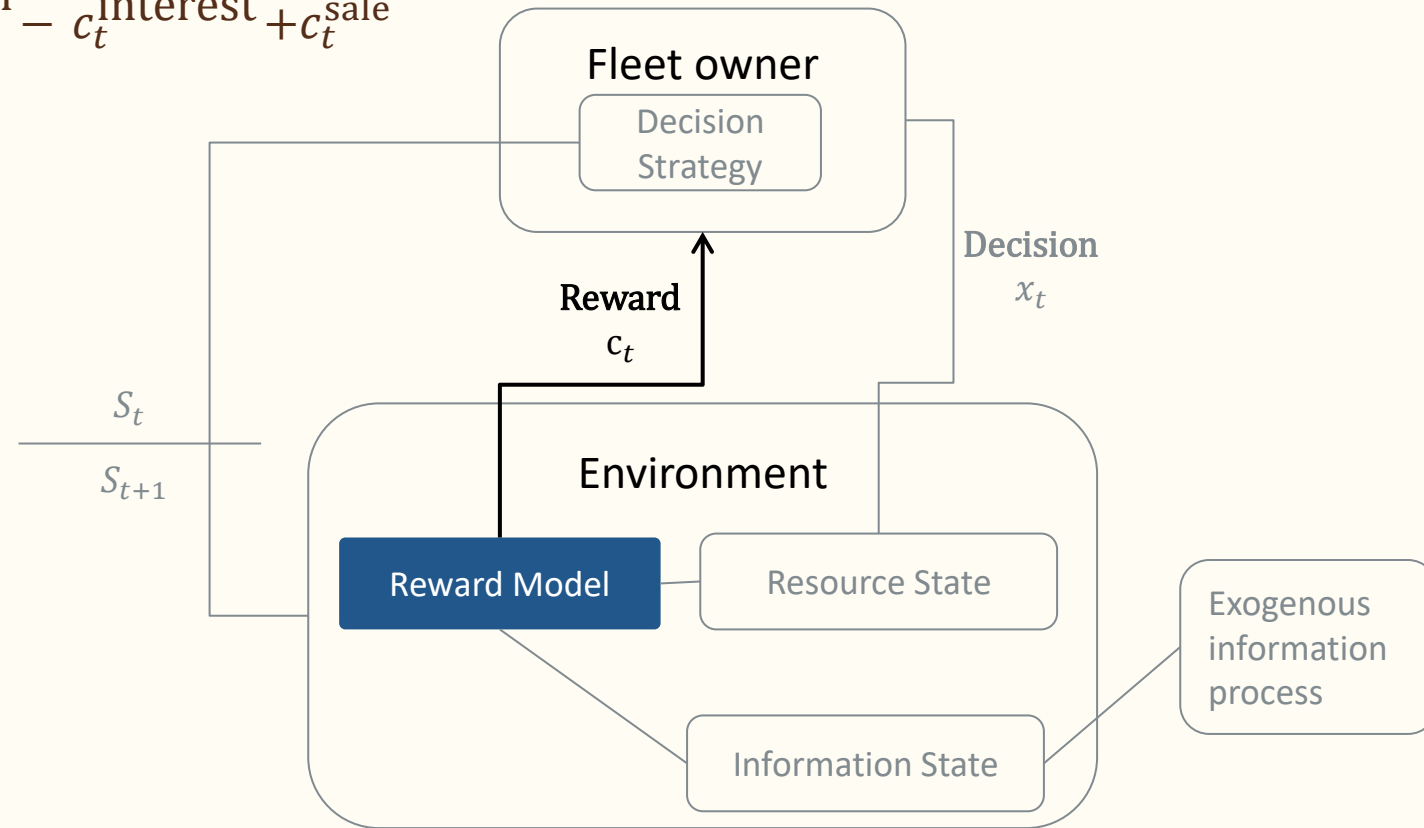
Reward model

$$C_t = c_t^{\text{revenue}} - c_t^{\text{energy}} - c_t^{\text{salary}} - c_t^{\text{depreciation}} - c_t^{\text{interest}} + c_t^{\text{sale}}$$

BET productivity



$$\max_{\pi} \mathbb{E} \left\{ \sum_{t=0}^T \gamma^t C(S_t, X^{\pi}(S_t)) \mid S_0 \right\}$$



Decision strategy

Lookahead optimization
(MILP) with rolling horizon

Forecasting profiles

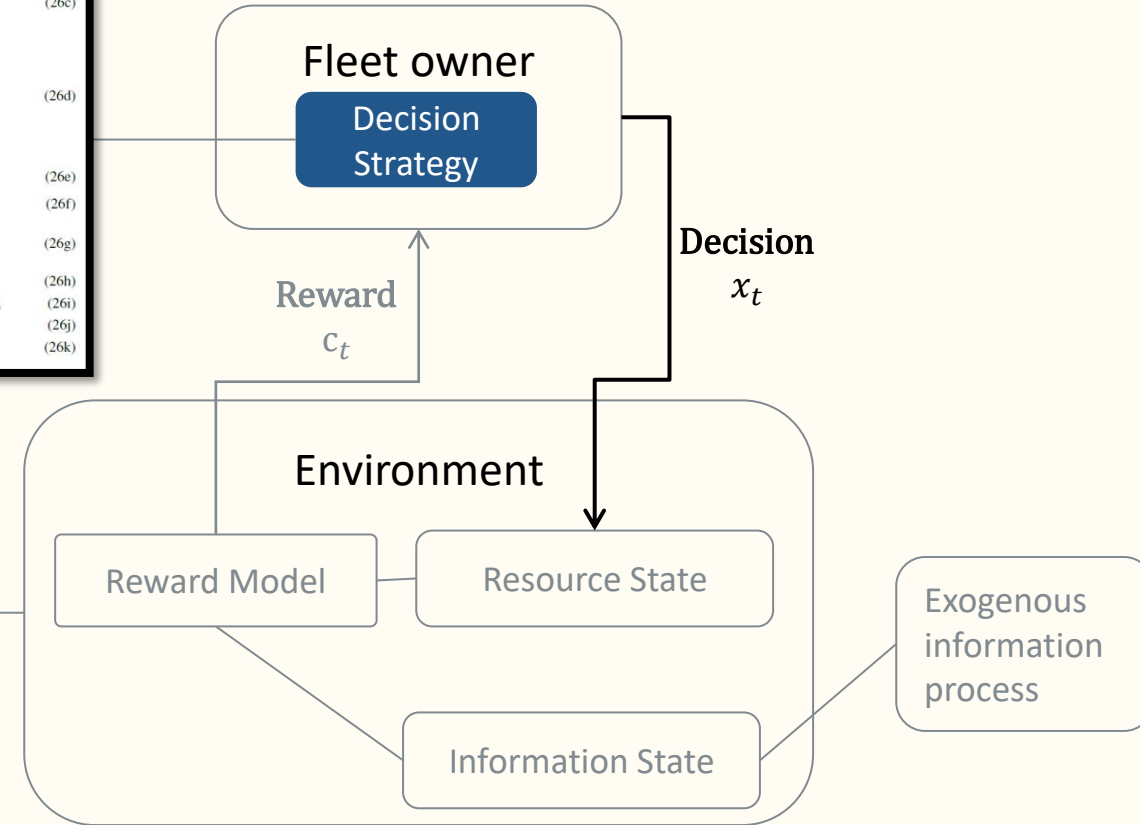
- No-change
- Low-progress
- High-progress

Reference strategy

- Myopic (maximize immediate reward)

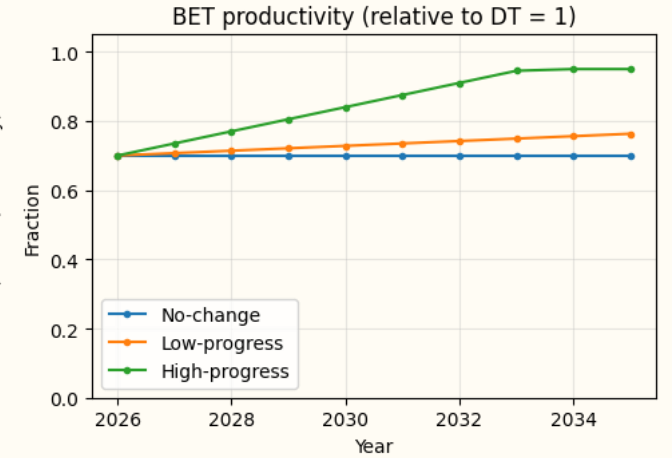
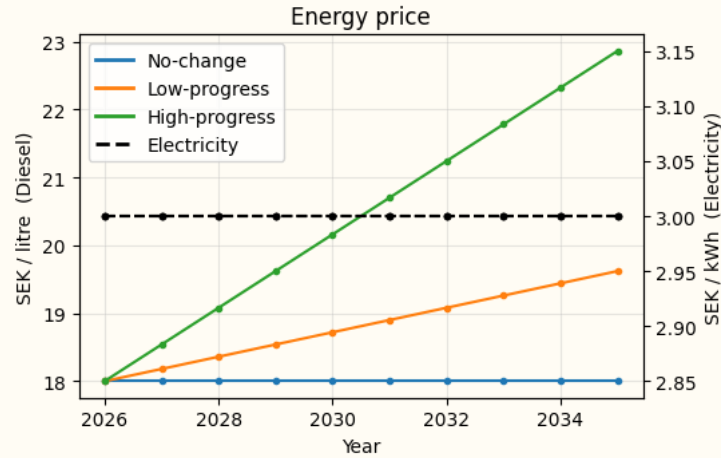
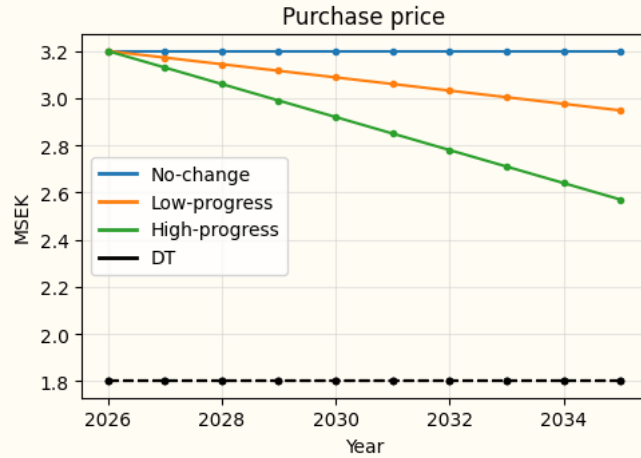
$$\begin{aligned}
 & \text{maximize}_{\tilde{x}_{i,t',j,s,d}} \sum_{t \in \mathcal{T}} \gamma^t \left\{ \underbrace{I \cdot M \sum_{i,s} \tilde{R}_{i,t',j,s} \cdot \rho(\tilde{v}, t')}_{\text{revenue}} - \underbrace{M \sum_{i,s} \tilde{R}_{i,t',j,s} \cdot \rho(\tilde{v}, t') \cdot K^e \cdot \tilde{p}_{i,t'}}_{\text{energy cost}} \right. & (26a) \\
 & \quad - \underbrace{L \sum_{i,s} \tilde{R}_{i,t',j,s} - \phi \cdot r \sum_{i,s} \tilde{R}_{i,t',j,s} \cdot \kappa(\tilde{v}, t') \cdot 1[\tilde{v}_a \leq A^{\max}]}_{\text{salary cost}} & (26b) \\
 & \quad - \underbrace{\sum_{i,s} \tilde{R}_{i,t',j,s} \cdot \frac{\kappa(\tilde{v}, t')}{A^{\text{econ}}} \cdot 1[\tilde{v}_a \leq A^{\text{econ}}]}_{\text{interest cost}} & (26c) \\
 & \quad \left. - \underbrace{\sum_{i,s,d \in \text{replace}^m} \tilde{x}_{i,t',j,s,d} \cdot (V^{\text{res}}(\tilde{v}, t') - V^{\text{book}}(\tilde{v}, t'))}_{\text{depreciation cost}} \right\} & (26d) \\
 & \quad + \underbrace{\sum_{i,s,d \in \text{replace}^m} \tilde{x}_{i,t',j,s,d} \cdot (V^{\text{res}}(\tilde{v}, t') - V^{\text{book}}(\tilde{v}, t'))}_{\text{sale result}} \\
 & \text{subject to} & \\
 & \quad \tilde{R}_{i,j,j,s} = 1 & \forall i \in \mathcal{I}, & (26e) \\
 & \quad \sum_{s \in \mathcal{V}} \tilde{R}_{i,t',j,s} = 1 & \forall i \in \mathcal{I}, t' \in \mathcal{T}, & (26f) \\
 & \quad \sum_{s \in \mathcal{V}, d \in \mathcal{D}} \tilde{x}_{i,t',j,s,d} = 1 & \forall i \in \mathcal{I}, t' \in \mathcal{T}, & (26g) \\
 & \quad \tilde{R}_{i,t',j,s} = 0 \Rightarrow \tilde{x}_{i,t',j,s,d} = 0, & & (26h) \\
 & \quad \tilde{x}_{i,t',j,s,\text{keep}} = 0 & \forall t', i, \tilde{v} : \text{age}(\tilde{v}) = A^{\max}, & (26i) \\
 & \quad \tilde{x}_{i,t',j,s,\text{keep}} = 1 \Rightarrow \tilde{R}_{i,t+1,j,s_{(m,a+1)}} = 1, & & (26j) \\
 & \quad \tilde{x}_{i,t',j,s,\text{replace}^m} = 1 \Rightarrow \tilde{R}_{i,t+1,j,s_{(m',0)}} = 1, & \forall m' \in \mathcal{M} & (26k)
 \end{aligned}$$

S_t
 S_{t+1}

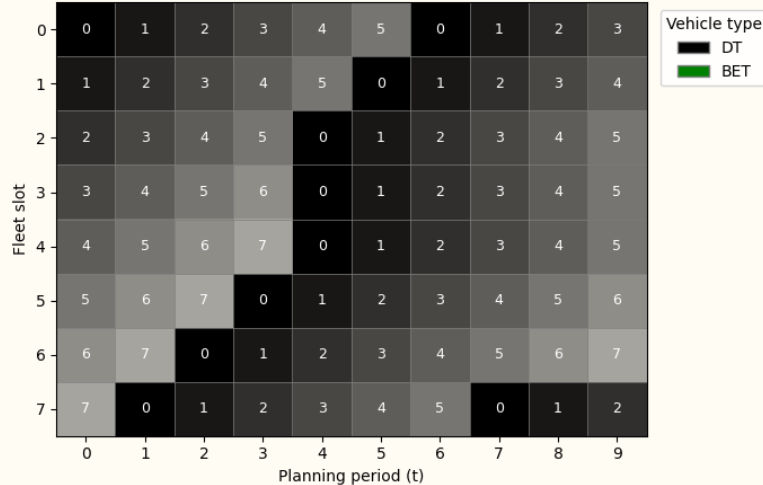


Forecasting profiles

Linear forecasts over the planning horizon (all profiles)

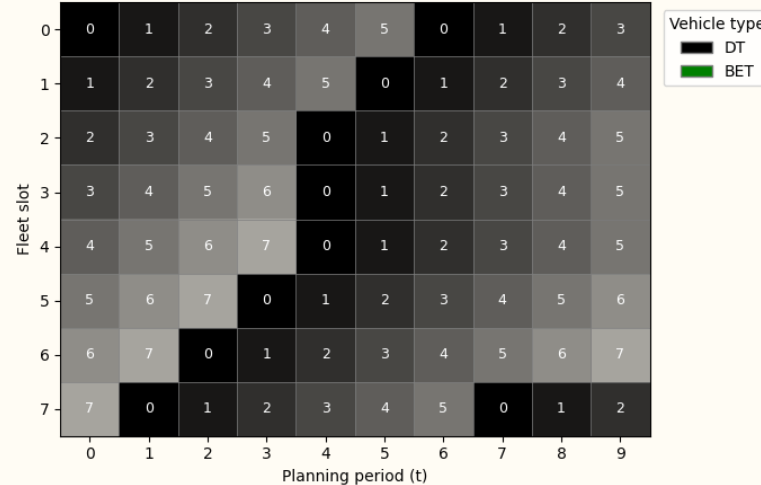


Fleet composition over the planning horizon (cell value = age)



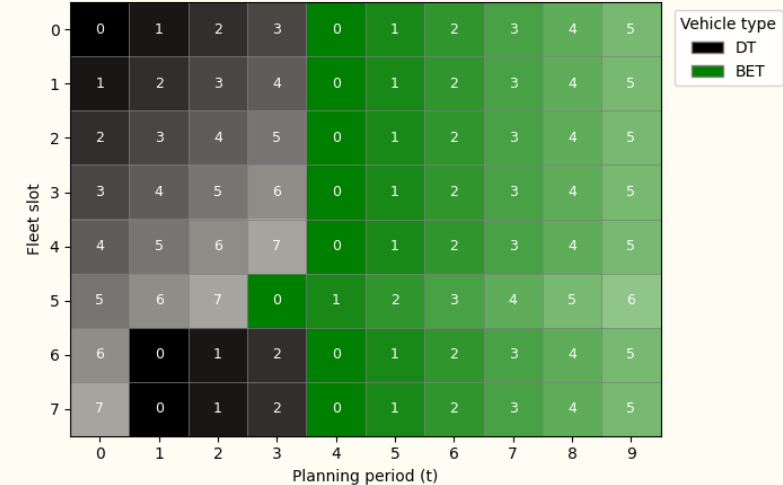
No-change

Fleet composition over the planning horizon (cell value = age)



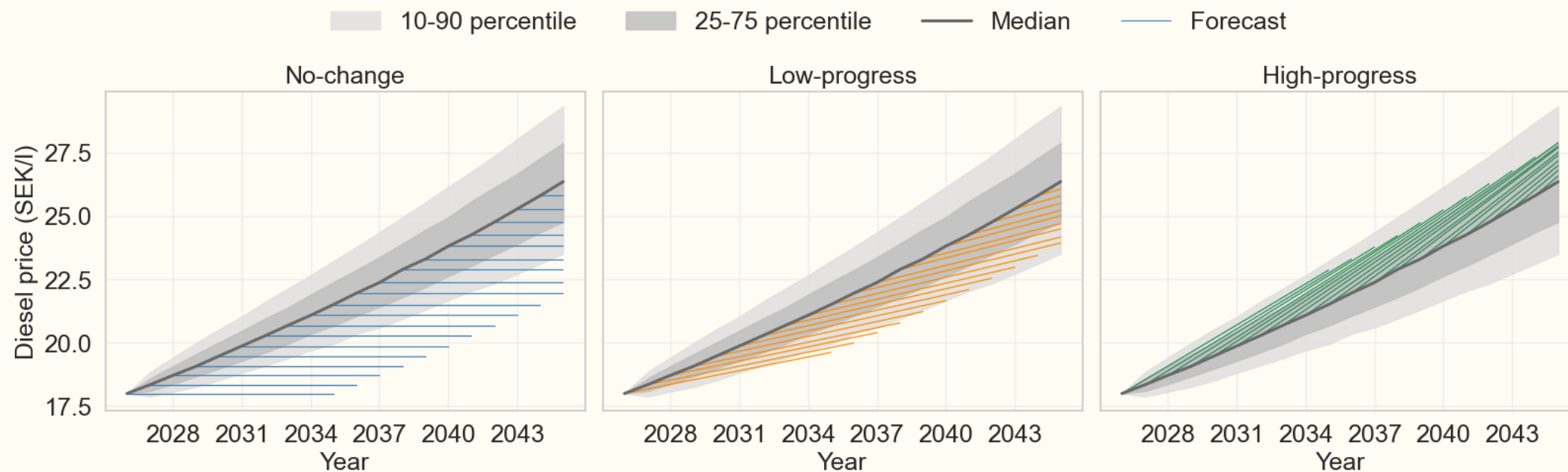
Low-progress

Fleet composition over the planning horizon (cell value = age)

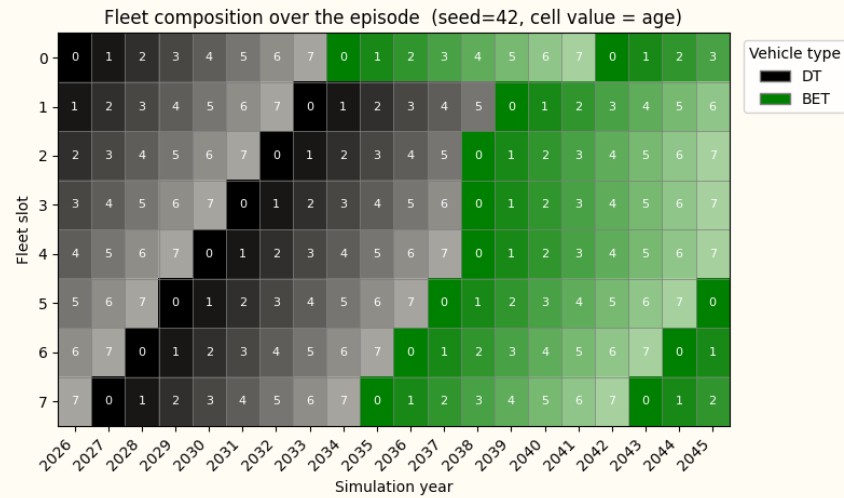


High-progress

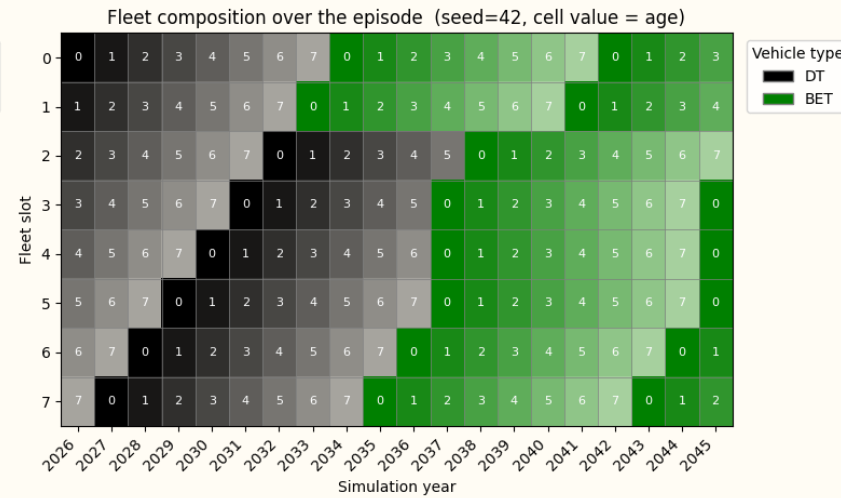
Forecasting profiles



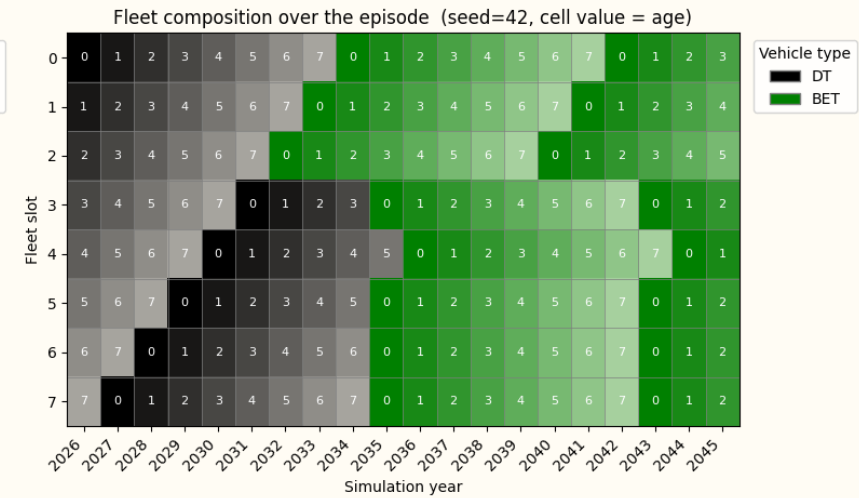
Rolling horizon



No-change

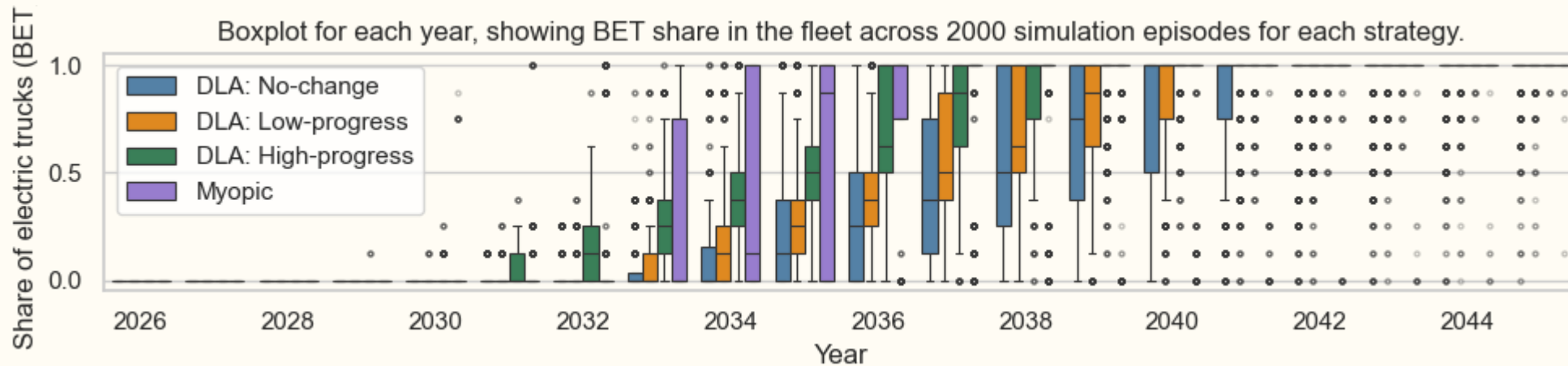
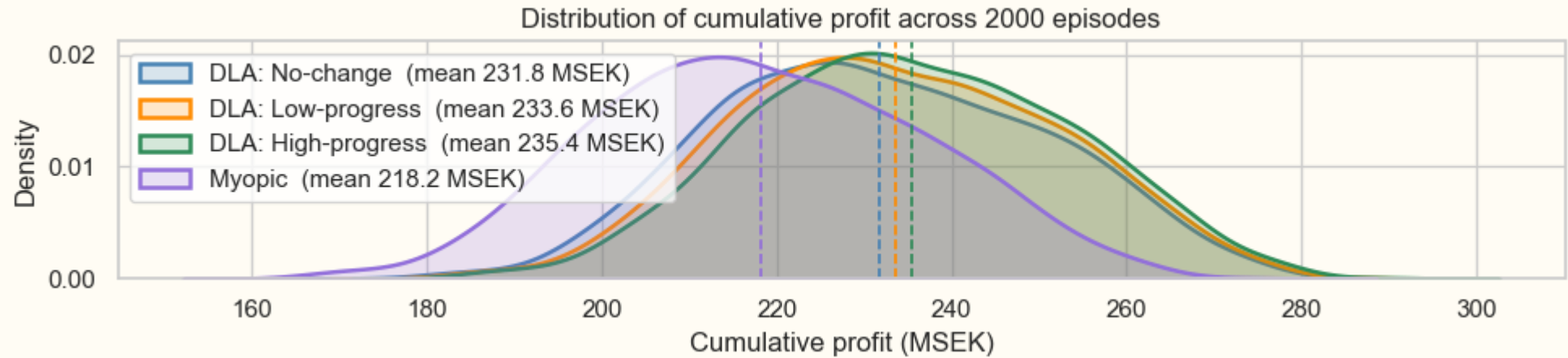


Low-progress



High-progress

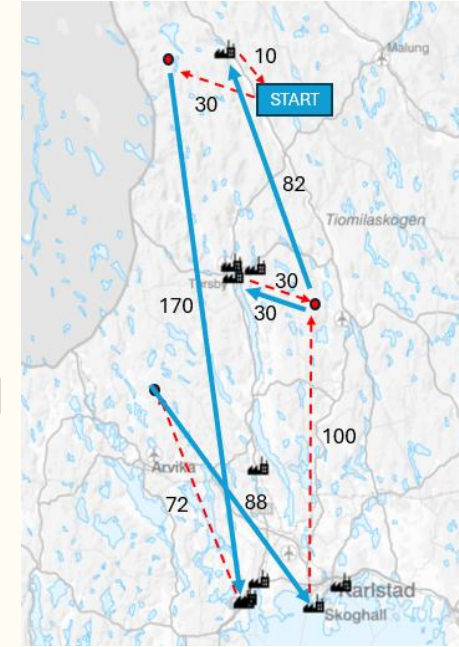
Results from 2000 simulations



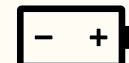
Conclusions & Future work

- Simulation model for the fleet replacement problem.
 - Lookahead optimization with rolling forecasts
 - Forecast profiles: *No-change, Low-progress, High-progress*
 - All lookahead variants outperform the myopic strategy but may have a slower electrification.
- Sequential decision framework
 - Path dependencies
 - Lock-in effects
 - Decision-maker behaviour
 - Winner and losers

- Split BET productivity variable into vehicle and infrastructure components
- Heterogenous fleet
- Integrate vehicle routing model (operational level planning)



- Include several BET variants



- Multi-actor simulation

